

A 16-Bit-Native Entropy Coding Pipeline for Lossless Medical Image Compression

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Abstract—Lossless compression is essential in medical imaging, where archival storage and diagnostic workflows require exact pixel reconstruction from 10–16-bit-per-sample data. Most entropy coders target 8-bit byte streams; medical images break this assumption by producing residual alphabets of thousands of symbols after spatial prediction. This paper presents MIC, a lossless codec for Digital Imaging and Communications in Medicine (DICOM) images that operates natively on 16-bit residuals. MIC combines spatial prediction, 16-bit run-length encoding, and large-alphabet Finite State Entropy (FSE), a table-driven implementation of asymmetric numeral systems (ANS). Rather than splitting 16-bit pixels into byte pairs before entropy coding, MIC treats each predicted residual as a single 16-bit symbol, capturing correlations between the high and low byte directly—a structural advantage quantified at 0.3–1.1 bits per symbol on the test dataset. The implementation extends FSE to support up to 65,535-symbol alphabets and includes a multi-state interleaved decoder that exploits instruction-level parallelism (ILP) during decompression. Evaluation on 21 de-identified DICOM images spanning nine modalities shows the 16-bit-native pipeline improves compression ratio by 5–22% (geometric mean 14%) over a Delta+Zstandard baseline, reaching a geometric-mean ratio of 3.12 times—within 1% of High-Throughput JPEG 2000 (HTJ2K, 3.15 times). A four-state decoder yields a 68% geometric-mean FSE stage decompression speedup on ARM64; strip-parallel configurations exceed HTJ2K decompression speed on all 21 images on both ARM64 and AMD64. A 20-kilobyte pure JavaScript decoder and a Go WebAssembly build enable client-side in-browser decoding, confirming that 16-bit-native entropy coding is a practical design point where throughput, simplicity, and deployment portability matter.

Index Terms—medical image compression, lossless compression, entropy coding, asymmetric numeral systems (ANS), finite state entropy (FSE), DICOM, high-bit-depth imaging.

I. INTRODUCTION

MEDICAL imaging systems routinely generate large volumes of high-bit-depth data. Modalities such as computed tomography (CT), magnetic resonance imaging (MR), digital radiography (computed radiography, CR; plain X-ray, XR), mammography, and tomosynthesis commonly store pixel values at 10, 12, or 16 bits per sample. A single digital breast tomosynthesis (DBT) acquisition view produces 60 or more reconstructed slices, totalling over 600 MB of raw pixel data per view. A complete bilateral four-view screening study exceeds 2 GB uncompressed. Efficient lossless compression is essential for archival storage, network transmission, and real-time rendering in clinical Picture Archiving and Communication Systems (PACS) and diagnostic viewers.

DICOM [1] supports several established lossless transfer syntaxes, including JPEG 2000, HTJ2K, JPEG-LS (a lossless/near-lossless predictive standard), and Run-Length Encoding (RLE) Lossless. These methods offer different tradeoffs among compression ratio, decoding speed, and implementation complexity [24], [25]. JPEG 2000 and HTJ2K provide strong compression performance but rely on transform and block-coding pipelines that are comparatively complex. JPEG-LS uses predictive coding and often yields strong lossless ratios, but its throughput characteristics remain application-dependent. RLE Lossless is simple but typically provides limited compression because it operates on byte-oriented runs and does not include an effective decorrelation stage.

This work studies a different design point for lossless medical image compression: direct coding of **predicted 16-bit residual streams**. Throughout this paper, a *residual stream* denotes the sequence of per-pixel prediction errors $r_{i,j} = x_{i,j} - \hat{x}_{i,j}$ produced by a spatial predictor, kept at the native 16-bit precision of the source pixels rather than re-serialised into bytes. As a concrete example: if the predicted pixel value is 1020 and the true pixel is 1023, the residual is +3. Smooth regions therefore produce many residuals near zero, and these near-zero residuals recur frequently—properties that motivate a 16-bit-native run-length and entropy coding stage.

The central observation is that the dominant entropy coders in modern compression systems, including Huffman coding, arithmetic coding, Lempel-Ziv 1977 (LZ77), and FSE/ANS as used in Zstandard, were designed primarily for 8-bit byte streams. Medical images break this assumption: DICOM stores pixels at 10–16 bits per sample, yielding symbol alphabets of 1,024 to 65,536 entries. Even after spatial delta prediction, the residual alphabet for a 16-bit CT image can span tens of thousands of distinct values. An entropy coder that assumes an 8-bit alphabet must either re-encode 16-bit symbols as byte pairs (losing inter-byte correlation and doubling the number of coding steps) or truncate the alphabet at 256 and fall back to raw storage for rare symbols. This mismatch is structurally costly: for example, small signed residuals such as $-2, -1, 0, +1$ have highly predictable high bytes, so encoding the high and low bytes independently forces the coder to spend bits on information already implied by the other byte. On our test images, the mutual information $I(X_H; X_L)$ between the high and low bytes of delta residuals ranges from 0.3 bits/symbol (MR) to 1.1 bits/symbol (MG3), corresponding to 8–22% of total entropy and explaining MIC’s 5–22% empirical advantage over Delta+Zstandard.

Based on this observation, we present MIC, a lossless codec that combines:

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- 1) a simple spatial predictor,
- 2) a 16-bit-native run-length representation of the residual stream, and
- 3) an extended FSE/tANS (table-based ANS) entropy coder supporting large active symbol alphabets.

The codec also includes multi-state interleaved entropy decoding to improve decompression throughput by increasing ILP.

A. Contributions

The main contributions of this paper are as follows.

- 1) **A 16-bit-native lossless coding pipeline for medical images.** We present a Delta+16-bit RLE+extended FSE pipeline for grayscale medical image compression that consistently outperforms Delta+Zstandard on all 21 tested images, with the improvement attributable to the structural cost of byte-splitting 16-bit residuals.
- 2) **An extended table-based entropy coder for large residual alphabets.** The proposed implementation supports up to 65,535 distinct symbols. One 16-bit code value ($2^{16}-1 = 65535$) is reserved as the overflow delimiter for full-range images, leaving up to 65,535 codes available for entropy coding. Tables are dynamically sized to the observed symbol range, reducing working-set size from 512 KB to approximately 2 KB for 8-bit images.
- 3) **A multi-state interleaved entropy decoder.** We implement and evaluate two-state and four-state FSE decoding to exploit ILP, achieving a geometric-mean +68% FSE stage speedup on ARM64 (+43% on AMD64) with no compression ratio penalty.
- 4) **An empirical evaluation against standard codecs and a general-purpose baseline.** We compare MIC with HTJ2K (OpenJPH) [29], JPEG-LS (CharLS) [30], and Delta+Zstandard on a dataset of 21 DICOM images spanning nine modalities (described in Section V).
- 5) **A compact JavaScript/WebAssembly (WASM) decoder implementation.** We provide a ~20 KB pure JavaScript (JS) decoder and a Go WebAssembly build to explore deployment feasibility for client-side viewing.

B. Scope

This paper is scoped to **lossless single-frame grayscale medical image compression**. Multi-frame and RGB extensions are outside the scope of this work. The present work does not address lossy coding, progressive decoding, or region-of-interest coding.

C. Organization

Section II reviews related work and positions MIC relative to existing methods. Section III describes the proposed coding pipeline. Section IV presents the multi-state entropy decoder. Section V describes experimental methodology. Section VI reports compression and speed results. Section VII concludes.

II. RELATED WORK

A. Lossless Compression in Medical Imaging

JPEG 2000 [2] employs the reversible 5/3 wavelet transform [20], [23] with Embedded Block Coding with Optimized Truncation (EBCOT), achieving excellent ratios at significant decompression overhead. HTJ2K [3], [27] replaces EBCOT with a faster block coder, substantially improving decode speed. JPEG-LS [4], [5] uses context-adaptive Median Edge Detector (MED) prediction with Golomb-Rice coding; it is an important lossless baseline as a standardized predictive codec widely deployed in medical PACS. RLE Lossless (DICOM TS 1.2.840.10008.1.2.5) applies byte-level PackBits without decorrelation, yielding poor ratios (typically $<1.5\times$). Higher-complexity approaches such as the Context-based Adaptive Lossless Image Codec (CALIC) [18] and the Free Lossless Image Format (FLIF) [19] achieve strong ratios on natural images but lack DICOM standardization and browser decoders.

These codecs are the primary baselines for this work. The goal is not to supersede them but to investigate a simpler residual-domain design specifically targeting high-bit-depth images with strong throughput and deployment portability.

B. Entropy Coding and the Byte Assumption

Asymmetric numeral systems (ANS) [7], [8] replace arithmetic coding's interval subdivision with integer state transitions. Finite State Entropy (FSE) [9] is a table-driven tANS implementation with $\mathcal{O}(1)$ per-symbol operations, popularized by Zstandard [10]. Recent work has formalized ANS efficiency bounds [12] and statistical interpretations [13]. In practice, Huffman coding becomes impractical above ~4,096 symbols and FSE in Zstandard is capped at 4,096; modern formats such as JPEG XL [6] likewise target byte streams.

Information-theoretic cost of byte-splitting. For a 16-bit symbol X decomposed into high byte X_H and low byte X_L , a byte-level coder achieves at best $H(X_H) + H(X_L) = H(X) + I(X_H; X_L)$. After delta prediction, when a residual is small (for example, ± 30), X_H is nearly deterministic given X_L , but it must still be encoded independently. On our test images, $I(X_H; X_L)$ ranges from 0.3 bits/symbol (MR) to 1.1 bits/symbol (MG3), corresponding to 8–22% of total entropy and closely matching MIC's empirical advantage over Delta+Zstandard.

C. ANS and Interleaved Decoding

Prior work has shown that interleaving or multi-stream organization can improve ANS decoder throughput. Giesen [14], [15] demonstrated the value of multi-stream range ANS (rANS); Lin *et al.* [16] extended parallel rANS to decoder-adaptive scalability; Weissenberger and Schmidt [17] showed massively parallel ANS decoding on GPUs. The present work does not claim novelty in ANS itself or in interleaving as a general concept. Rather, the contribution lies in applying these ideas in a 16-bit-native medical image codec, together with a specific large-alphabet implementation and throughput-oriented software design.

D. Positioning of This Work

Table I distinguishes MIC from prior work across five dimensions relevant to medical image compression.

TABLE I: Feature comparison of MIC with related codecs and entropy coding methods.

Feature	JPEG-LS	HTJ2K	Zstd	rANS	MIC
Lossless medical	Yes	Yes	General	General	Yes
Native 16-bit	No	No	No	Configurable	Yes
Multi-state ILP	No	No	No	Yes	Yes
Browser decoder	No	No	Partial	No	Yes
Source lines	5k	20k	N/A	N/A	1.1k

III. PROPOSED METHOD

Each stage of the pipeline described in Section I is simple enough to decode in a single sequential pass with minimal branching, as illustrated in Fig. 1.

A. Spatial Prediction

Let $x_{i,j}$ denote the pixel at row i , column j . For interior pixels, MIC predicts the current pixel from the average of the top and left neighbors:

$$\hat{x}_{i,j} = \left\lfloor \frac{x_{i-1,j} + x_{i,j-1}}{2} \right\rfloor \quad (1)$$

where the division is integer division rounded down (matching the decoder exactly), and all arithmetic is performed in unsigned integers. The residual is formed as:

$$r_{i,j} = x_{i,j} - \hat{x}_{i,j}. \quad (2)$$

Boundary pixels use only the available neighbor, and the first pixel is stored directly. This predictor was selected because it is computationally simple, branch-light in decoding, and effective on the current dataset.

Predictor comparison. We implemented and compared four predictors through the full RLE + FSE pipeline on the 21-image dataset: left-only (left neighbor, no top), average (MIC default), Paeth (PNG predictor [26]), and MED (JPEG-LS [4]). Table II summarizes geometric means and relative gains.

TABLE II: Predictor ablation summary: geometric means and win counts across all 21 images.

Predictor	Geo. Mean	Wins	Avg→X
Left-only	3.38×	3/21	−2.3%
Average (MIC)	3.46×	13/21	baseline
Paeth	3.48×	2/21	+0.5%
MED (JPEG-LS)	3.52×	16/21	+1.6%

MED decompression is 1.5–2× slower due to the three-way conditional branch and diagonal neighbor dependency that prevents the branch-free interior loop used by the average predictor. MED yields the best geomean ratio (+1.6% over Avg) but at significant speed cost; the average predictor is retained as the default for its consistent per-modality performance and branch-free decompression path.

B. Effective Bit Depth and Overflow Coding

For 16-bit images, residuals can span $[-65535, +65535]$. Rather than widening the main residual stream to 32 bits, MIC uses an overflow delimiter scheme derived from the effective image bit depth:

$$d = \lceil \log_2(\max(x) + 1) \rceil. \quad (3)$$

Residuals within an in-range interval are mapped directly to 16-bit codes. Residuals outside that interval are encoded using a delimiter followed by the raw pixel value.

Encoding procedure:

- 1) Compute $\text{deltaThreshold} = 2^{d-1} - 1$.
- 2) Compute delimiter $D = 2^d - 1$.
- 3) For each residual $r_{i,j}$: if $|r_{i,j}| < \text{deltaThreshold}$, emit `uint16(deltaThreshold +  $r_{i,j}$ )`; otherwise emit delimiter D , then emit raw pixel $x_{i,j}$.

The threshold condition is strict ($<$, not $\leq$), so residuals with $|r| = \text{deltaThreshold}$ take the overflow path. Residual zero maps to code value `deltaThreshold` exactly. Table III illustrates the mapping for a representative bit depth.

TABLE III: Overflow coding example for $d = 12$ bits ($\text{deltaThreshold} = 2047$, $D = 4095$).

Residual r	Emitted code(s)	Notes
−2	2045	in-range
−1	2046	in-range
0	2047	in-range (= threshold)
+1	2048	in-range
+2	2049	in-range
±2047	4095, raw pixel	overflow

Non-collision proof. In-range residuals satisfy $|r| \leq \text{deltaThreshold} - 1$, so stored values range from 1 to $2^d - 3$. The delimiter is $D = 2^d - 1$. Since $2^d - 3 < 2^d - 1$, the stored value of a direct residual can never equal D ; the two ranges are disjoint.

Decoding procedure: Read the next `uint16` value c . If $c \neq D$, recover the residual as $r = \text{int16}(c - \text{deltaThreshold})$ and reconstruct $x_{i,j} = \hat{x}_{i,j} + r$. If $c = D$, the raw pixel follows: read the next `uint16` as $x_{i,j}$ directly.

C. 16-Bit Run-Length Encoding

The predicted residual stream is converted into an alternating sequence of:

- **Same runs:** a count and a repeated 16-bit value;
- **Diff runs:** a count followed by a sequence of distinct 16-bit values.

A same run is emitted only when at least three consecutive identical values are observed.

Midpoint protocol. Run headers are stored as `uint16` values. A midpoint constant `midCount` = 32767 partitions the header space: a count $c \leq \text{midCount}$ signals a same run of length c ; a count $c > \text{midCount}$ signals a diff run of length $c - \text{midCount}$. The maximum same-run or diff-run length is therefore 32767 symbols; longer sequences are split into multiple headers transparently. The decoder distinguishes the two run types solely by comparing the header value against `midCount`.

MIC Compression Pipeline



Fig. 1: MIC compression pipeline. Raw 16-bit pixels are delta-encoded (avg of top and left neighbors), run-length encoded (same/diff runs), and entropy-coded with a 16-bit-native FSE/tANS coder.

Encoding example. For the residual code sequence $[5, 5, 5, 5, 9, 10, 11, 11, 11]$, the RLE output is: same($c = 4$, value=5), diff($c = 2$, values=9,10), same($c = 3$, value=11). The encoded stream contains five uint16 words: $[4, 5, 32769, 9, 10, 3, 11]$, where $32769 = \text{midCount} + 2$ signals a diff run of length 2.

Why 16-bit RLE matters. A run of 1,000 identical 16-bit residuals $v = 0x0103$ is encoded by MIC as two uint16 words (a count and the value), regardless of v 's byte pattern. A byte-oriented RLE sees the interleaved stream $0x03, 0x01, 0x03, 0x01, \dots$: no single byte repeats 1,000 times, so it must emit two interleaved runs or fall back to literals.

Same-run fast path. Same-value runs (often $>80\%$ of all runs on smooth medical images after delta coding) are fast-pathed in the decoder to return the cached value without touching the compressed-data slice. Each output symbol in a same run requires only a counter decrement.

D. Large-Alphabet FSE Coding

FSE overview. FSE maintains a single integer *state*. To decode one symbol, the decoder indexes a prebuilt decode table using the current state. Each table entry stores the output symbol, the number of bits n to read from the bitstream, and a base value b for the next state. The decoder reads n bits from the bitstream and computes the next state as $b+$ (bits read). This requires only a table lookup, a bit read, and an addition per symbol; no divisions or multiplications are needed. **Encoding** processes symbols in reverse order (last symbol first), building the compressed bitstream from the end; **decoding** reads bits forward.

The RLE output is entropy-coded using table-based ANS (tANS), implemented as Finite State Entropy. MIC extends FSE to support up to 65,535 distinct symbols. As noted in Section I, one 16-bit code value is reserved as the overflow delimiter, leaving at most 65,535 codes available for residual symbols. This contrasts with the 4,096-symbol limit in Zstandard's FSE implementation, which was designed for byte-oriented streams.

Dynamic table sizing. Encode and decode tables are sized to the actual observed symbol range rather than the theoretical maximum. For 8-bit images stored in 16-bit containers

(common in MR), this reduces the working set from 512 KB to approximately 2 KB.

E. Adaptive Table Size Selection

The FSE *tableLog* parameter determines the size of the decode table (2^{tableLog} entries) and the precision of symbol probability estimates. MIC automatically selects *tableLog* based on the number of active symbols and observation density:

- *symbolLen* = number of distinct symbols observed in the RLE output stream (equivalently, the observed symbol range: max value minus min value plus 1, counting all positions in that range even if some go unused).
- *inputLength* = total number of symbols in the RLE output stream fed to the FSE coder.
- *symbolDensity* = $\text{inputLength} / \text{symbolLen}$, i.e., average observations per symbol slot.

Selection rules (applied in order; later rules override earlier ones):

- 1) Default: *tableLog* = 11.
- 2) If *symbolLen* > 128 **and** *symbolDensity* > 32 : set *tableLog* = 12.
- 3) If *symbolLen* > 512 **and** *symbolDensity* > 16 : set *tableLog* = 13 (overrides rule 2 when both conditions hold).

A larger *tableLog* improves probability precision for the FSE model but increases table memory from $2^{11} \times 4 = 8$ KB to $2^{13} \times 4 = 32$ KB. Rule 2 activates on images with a large active symbol set and enough data per symbol for reliable probability estimates. Rule 3 extends this to images with very broad symbol ranges (e.g., mammography after delta coding), where precision gains outweigh the memory cost even at lower density.

Table IV presents the *tableLog* ablation across selected images.

Nine images benefit from TL=12 or TL=13 (CR +6.3%, MG1 +9.9%, MG2 +9.9%, MR2 +4.6%, RG2 +9.9%, RG3 +7.8%, XA1 +4.4%, MR3 +0.9%, NM1 +1.9%); the adaptive policy correctly identifies all beneficial cases. Decompression throughput is unaffected by *tableLog* choice ($<5\%$ variation).

TABLE IV: TableLog ablation: forced TL=11/12/13 vs. adaptive selection (selected images, Apple M2 Max).

Image	TL=11	TL=12	TL=13	Adaptive
CR	3.47×	3.63×	3.69×	3.69×
MG1	8.00×	8.57×	8.79×	8.79×
MG2	7.98×	8.55×	8.77×	8.77×
MR2	3.14×	3.24×	3.28×	3.28×
RG2	3.85×	4.12×	4.23×	4.23×
RG3	5.64×	5.95×	6.08×	6.08×

IV. MULTI-STATE ENTROPY DECODING

A. The Serial Dependency Problem

A conventional single-state table-based ANS decoder contains a serial dependency chain. Each decoded symbol requires one table lookup followed by one bit read, and the next state depends on both results:

$$\begin{aligned} e &= \text{decTable}[s_n], \\ s_{n+1} &= e.\text{base} + \text{getBits}(e.\text{nbBits}), \end{aligned} \quad (4)$$

where s_n is the current decoder state, decTable is the prebuilt decode table, $e.\text{base}$ is the base value for the next state stored in the table entry, $e.\text{nbBits}$ is the number of bits to consume from the bitstream, and $\text{getBits}(k)$ reads the next k bits from the compressed stream and returns them as an integer. Because s_{n+1} depends on e , which depends on s_n , the computations for symbol $n+1$ cannot begin until symbol n is fully resolved. With a table-lookup latency of approximately $L \approx 4$ cycles on modern out-of-order CPUs (L1 cache hit), the loop is limited to roughly one symbol per L cycles regardless of available execution units. Modern CPUs have 4–6 execution ports; a single-state ANS decoder uses exactly one dependency chain, leaving 75–83% of the hardware capacity idle.

B. Interleaved Decoding

MIC breaks this dependency by running multiple independent state machines on interleaved symbol positions. In the four-state design, four independent chains (A, B, C, D) reconstruct positions:

chain A : 0, 4, 8, ...
chain B : 1, 5, 9, ...
chain C : 2, 6, 10, ...
chain D : 3, 7, 11, ...

The four chains are completely independent: each has its own state variable, its own sequence of table lookups, and its own bit-extraction operations. The CPU’s out-of-order engine sees four simultaneous table-lookup address streams.

Bitstream format. The encoder partitions the output symbol sequence into four interleaved subsequences, encodes each independently (producing four FSE-compressed streams), and concatenates them. The four stream lengths are stored as a 4-entry header so the decoder can locate each stream’s start offset. All four chains use the same FSE decode table (built from the joint histogram over all symbols). Initial state values for each chain are stored as part of the per-chain header. Output reordering is performed by the decoder, which writes

reconstructed symbols to positions $4k$, $4k+1$, $4k+2$, $4k+3$ from chains A, B, C, D respectively.

Latency model. Let L be the table-lookup latency (cycles). Single-state FSE processes 1 symbol per L cycles. The k -state decoder processes k symbols per L cycles (amortized):

TABLE V: Multi-state decoder latency model and empirical speedup.

States (k)	Amortized latency	Theoretical	Empirical
1	L cy/sym	1.0×	1.0×
2	$L/2$ cy/sym	2.0×	1.3×
4	$L/4$ cy/sym	4.0×	2.0×

The empirical speedup is below theoretical maximum due to: (1) bit-reader refill overhead shared across all chains, (2) instruction cache pressure from the unrolled loop, and (3) memory bandwidth limits on compressed stream reads [28].

C. Correctness and Signaling

Correctness follows from partitioning the output sequence into independent subsequences during encoding and reversing that assignment during decoding; each chain operates on a strict subset of positions with no cross-chain data dependency.

The bitstream signals the decoding mode with a 2-byte magic header: $0xFF, 0x02$ for two-state; $0xFF, 0x04$ for four-state; followed by a 4-byte symbol count. In the single-state format, the first byte encodes `tableLog` (the value that determines the decode table size as 2^{tableLog} entries).

The `tableLog` is bounded by the symbol alphabet size: for a 16-bit symbol alphabet of at most 65,535 symbols, a table of $2^{16} = 65,536$ entries gives one slot per symbol, which is the practical minimum for meaningful probability quantization. The implementation supports `tableLog` up to 17 (giving 131,072 entries, approximately $2\times$ the alphabet size, for better probability resolution on large alphabets). Since the maximum valid `tableLog` value is 17 and a single byte can represent at most 255, the value $0xFF (= 255 > 17)$ is not a valid single-state `tableLog` and can safely be reused as the extended-format marker. All three formats coexist without ambiguity.

Platform-specific kernels. On AMD64, Bit Manipulation Instruction Set 2 (BMI2) `SHLQ/SHRQ` instructions provide 3-operand variable shifts without contention on the `CL` register (the x86 shift count register used by traditional variable shifts); on ARM64, `LSLV/LSRV` (logical shift left/right variable) serve the same role natively. Pure-Go implementations cover all other targets.

D. Multi-State FSE Decoder Implementation

zeroBits flag. The FSE decoder uses a fast inner loop that reads a fixed number of bits per symbol. However, when any symbol’s normalized count exceeds half the table size (i.e., its probability exceeds 50%), some state transitions must emit zero bits—the decoder would call `getBits(0)`. A boolean flag named `zeroBits` (set at table-build time in the implementation) tracks this condition; Pure-Go code must branch on it to avoid a zero-shift undefined behavior, and the resulting conditional branch breaks the fast path. The AMD64 BMI2

assembly function `fse4StateDecompKernel` (described below) uses the `SHRXQ` instruction, which handles a shift count of zero correctly without branching, so it sidesteps the `zeroBits` check entirely.

`fse4StateDecompKernel` is a hand-written AMD64 assembly routine in the MIC implementation that runs all four interleaved FSE decode chains in a single tight loop. It uses BMI2 `SHLXQ/SHRXQ` instructions for register-saving variable shifts (avoiding contention on the CL shift-count register used by legacy x86 variable-shift instructions) and issues four simultaneous table-lookup address computations to keep the CPU's out-of-order execution engine fully occupied.

V. EXPERIMENTAL METHODOLOGY

A. Dataset

The evaluation uses 21 DICOM images spanning nine modalities, drawn from publicly available datasets: the NEMA Working Group 4 (WG-04) compression sample images [31], the NEMA 1997 CD, and the Clunie Breast Tomosynthesis Case 1 (Table VI).

TABLE VI: Test dataset: 21 clinical DICOM images spanning nine modalities.

Image	Modality	Dimensions	Raw (MB)
MR	Brain MRI	256×256	0.13
CT	CT scan	512×512	0.50
CR	Comp. radiography	2140×1760	7.18
XR	X-ray	2048×2577	10.1
MG1	Mammography	2457×1996	9.35
MG2	Mammography	2457×1996	9.35
MG3	Mammography	4774×3064	27.3
MG4	Mammography	4096×3328	26.0
CT1	CT scan	512×512	0.50
CT2	CT scan	512×512	0.50
MG-N	Mammography	3064×4664	27.3
MR1	Brain MRI	512×512	0.50
MR2	Brain MRI	1024×1024	2.00
MR3	Brain MRI	512×512	0.50
MR4	Brain MRI	512×512	0.50
NM1	Nuclear med.	256×1024	0.50
RG1	Radiography	1841×1955	6.86
RG2	Radiography	1760×2140	7.18
RG3	Radiography	1760×1760	5.91
SC1	Sec. capture	2048×2487	9.71
XA1	Fluoroscopy	1024×1024	2.00

This dataset spans multiple modalities and image sizes but remains modest for strong generalization claims.

B. Baselines

The current study compares MIC with:

- **HTJ2K** (OpenJPH [29]),
- **JPEG-LS** (CharLS [30]), and
- **Delta + Zstandard** (level 19).

All three baselines are called in-process (no subprocess or file-I/O overhead) using Go's C foreign-function interface (CGO) for the C-library codecs.

C. Metrics

The paper reports:

- **Compression ratio:** uncompressed size / compressed size.

- **Decompression throughput:** uncompressed pixel bytes reconstructed per second (MB/s).
- **Encoding throughput:** uncompressed pixel bytes processed per second (MB/s).

D. Benchmark Procedure

All codecs were measured using in-process library calls on the same machine to avoid file-I/O and subprocess overhead. The Go testing framework's `-benchtime=10x` flag was used (10 iterations per benchmark; each iteration processes the full image). Run-to-run variance is <2% on Apple M2 Max and <5% on Intel AMD64.

Platforms and compiler flags:

- Apple M2 Max (12-core ARM64), macOS. Go compiler (gc) with default optimization; CGO-linked C components compiled with `-O3`.
- Intel Core Ultra 9 285K (x86-64/AMD64, mixed P-core/E-core topology), Linux. Go compiler (gc) with default optimization; C components compiled with `-O3 -march=native`.

Codec variants: All variants produce the same compressed bytes unless noted. Differences are in the decoder implementation only.

- **MIC-Go:** Pure Go, single-state FSE decoder (Section III).
- **MIC-4state:** Pure Go, four-state interleaved FSE decoder (Section IV).
- **MIC-4state-C:** C implementation of the four-state FSE decoder, called via CGO. Produces bit-identical output to MIC-4state.
- **MIC-4state-SIMD:** C four-state FSE decoder with the BMI2 `fse4StateDecompKernel` on AMD64; falls back to the scalar C decoder on ARM64 (ARM64 lacks BMI2). Bit-identical to MIC-4state-C.
- **Wavelet+SIMD:** Le Gall 5/3 integer wavelet (blocked column layout for cache efficiency) with AVX2 predict/update kernels on AMD64. On ARM64 the same blocked column layout is used without SIMD (*scalar blocked*): the cache-friendly access pattern still improves throughput over a plain column-by-column transform even without vectorization. Produces a different compressed stream from all MIC variants.
- **PICS-C-N:** Parallel Image Compressed Strips. The image is split into N equal horizontal strips; each strip is compressed independently using its own FSE table. Decompression of all strips runs in parallel using a C pthreads worker pool; strips are independently decodable. $N \in \{2, 4, 8\}$ in this evaluation.

E. JavaScript and Browser Evaluation

A pure JavaScript decoder (~20 KB ES module, zero Node Package Manager (npm) dependencies) and a Go WebAssembly build (~2.5 MB) were implemented. Performance measurements reported in Section VI were obtained in Node.js v24.8 (Apple M2 Max).

VI. RESULTS

A. Compression Ratio

Table VII presents lossless compression ratios for all codec variants across the 21-image dataset.

Analysis. JPEG-LS achieves the highest lossless ratio on all 21 images, as expected given its context-adaptive MED predictor. MIC’s design target is not ratio alone but the ratio-throughput-simplicity tradeoff: MIC achieves approximately 91% of JPEG-LS’s geometric mean ratio ($3.12\times$ vs. $3.44\times$) while decoding faster and requiring approximately 1/5 the source lines. Mammography achieves the highest MIC ratios (up to $8.79\times$) due to smooth tissue regions producing near-zero RLE runs.

PICS strip adaptation. Each PICS strip receives a dedicated FSE table that specializes to its local residual distribution. On large images, this per-strip specialization improves ratio slightly. Formally, if S_k denotes the RLE symbol sub-sequence for strip k , $|S_k|$ its length, and $H(S_k)$ its Shannon entropy (bits per symbol), then

$$\sum_k |S_k| \cdot H(S_k) \leq |S| \cdot H(S), \quad (5)$$

where S is the full sequence and the inequality is strict when residual statistics differ across strips. In plain terms: a table that is specialized to a strip’s own pixel-value distribution is at least as efficient as a single global table, and strictly better when different image regions have different statistical properties.

B. Encoding Throughput

Tables VIII and IX report per-image encoding throughput for all codec variants on AMD64 and ARM64 respectively. On ARM64, MIC-4state-C is $2\text{--}2.5\times$ faster to encode than pure-Go MIC-Go; JPEG-LS is consistently $2\text{--}4\times$ slower than MIC-Go. PICS-8 reaches $1.2\text{--}2.1$ GB/s on large images, sufficient for real-time PACS ingestion.

C. Decompression Throughput—ARM64

Table X presents decompression throughput on Apple M2 Max (ARM64). On ARM64, all MIC variants use pure-Go or C scalar implementations; there is no SIMD column. MIC-4state-C is the fastest single-threaded variant on 13 of 21 images; PICS-C-8 exceeds HTJ2K decompression throughput on all 21 images.

[†]MR is too small for PICS-C-8; PICS-C-4 is best. MIC-4state-C is $1.7\text{--}5.0\times$ faster than JPEG-LS on all 21 images. PICS-C-8 peaks at 3,773 MB/s on MG2.

D. Decompression Throughput—AMD64

Table XI presents decompression throughput on Intel Core Ultra 9 285K (AMD64). On AMD64, MIC-4state-SIMD uses the BMI2 `fse4StateDecompKernel` assembly routine, and Wavelet+SIMD uses AVX2 predict/update kernels. MIC-4state-SIMD is the fastest single-threaded variant on 16 of 21 images; PICS-C-8 exceeds HTJ2K decompression throughput on all 21 images on both platforms.

[†]MR is too small for PICS-C-8; PICS-C-4 achieves the highest PICS throughput on this image. HTJ2K leads single-threaded on MG1, MG2, MG4, CT, and RG3.

E. FSE Stage Microbenchmarks

Table XII reports FSE decompression throughput measured in isolation (MB/s over the uncompressed RLE symbol stream), decoupling the entropy-coding stage from delta and RLE overhead, for a representative subset of modalities on both platforms.

*CR single-state is depressed by the zeroBits branch (see Section IV); multi-state interleaving amortizes this overhead across four chains. The AMD64 `fse4StateDecompKernel` sidesteps the branch entirely, yielding +127%.

The 4-state decoder delivers a geometric-mean speedup of +68% on ARM64 and +43% on AMD64 over the 1-state baseline. The smaller AMD64 gain reflects a higher 1-state baseline: the Intel out-of-order engine tolerates the serial dependency chain more efficiently, compressing the relative multi-state improvement.

F. JavaScript Runtime Results

Table XIII reports Node.js single-threaded performance; Table XIV reports parallel PICS decoding via `worker_threads`.

MG1 (9.35 MB) decompresses in 19.4 ms at 483 MB/s using 12 workers. A web-based DICOM viewer can fetch a `.mic` file directly from object storage and decode it client-side without server-side compute or transcoding latency.

Limitation. These measurements were obtained in Node.js `worker_threads`, which does not fully replicate browser Web Worker performance. Full browser performance benchmarking remains future work.

VII. CONCLUSION

This paper presented MIC, a lossless medical image compression pipeline based on spatial prediction, 16-bit-native run-length representation, and large-alphabet table-based entropy coding. The key results across 21 clinical DICOM images spanning nine modalities:

- **Compression ratios:** $1.7\times\text{--}8.9\times$ across the 21 evaluated grayscale images.
- **vs. Delta+Zstandard:** 5–22% better compression on all 21 images (geometric mean +14%), attributable to the structural cost of byte-splitting 16-bit residuals.
- **vs. HTJ2K:** MIC-4state-C exceeds HTJ2K decompression on 19/21 images single-threaded (ARM64) and 16/21 (AMD64); PICS-C-8 exceeds HTJ2K on all 21 images on both platforms.
- **vs. JPEG-LS:** $1.7\text{--}5.0\times$ faster decompression; JPEG-LS retains the best lossless ratios (geometric mean $3.44\times$ vs. MIC’s $3.12\times$).
- **Multi-state decoder:** geometric-mean +68% FSE stage speedup on ARM64 (+43% AMD64) with no ratio penalty.
- **Browser decoder:** ~20 KB pure JS decoder; PICS with `worker_threads` achieves 483 MB/s on MG1 in Node.js.
- **Implementation:** approximately 1,100 lines of Go code, about $18\times$ fewer source lines than HTJ2K.

TABLE VII: Lossless compression ratios: all codec variants, 21 images. Bold = best ratio per row.

Image	Raw (MB)	Δ +Zstd-19	MIC	Wavelet	PICS-4	PICS-8	HTJ2K	JPEG-LS
MR	0.13	1.95×	2.35×	2.38×	2.28×	2.21×	2.38×	2.52×
CT	0.50	2.05×	2.24×	1.67×	2.15×	1.96×	1.77×	2.68×
CR	7.18	3.24×	3.69×	3.81×	3.70×	3.71×	3.77×	3.96×
XR	10.1	1.43×	1.74×	1.76×	1.75×	1.76×	1.67×	1.76×
MG1	9.35	7.20×	8.79×	8.67×	8.84×	8.87×	8.25×	8.91×
MG2	9.35	7.21×	8.77×	8.65×	8.83×	8.85×	8.24×	8.90×
MG3	27.3	2.04×	2.24×	2.32×	2.31×	2.34×	2.22×	2.38×
MG4	26.0	3.10×	3.47×	3.59×	3.59×	3.62×	3.51×	3.71×
CT1	0.50	2.47×	2.79×	2.49×	2.54×	2.29×	2.70×	3.19×
CT2	0.50	3.24×	3.49×	2.87×	3.11×	2.72×	3.29×	4.54×
MG-N	27.3	2.04×	2.24×	2.32×	2.31×	2.34×	2.23×	2.38×
MR1	0.50	1.79×	2.09×	2.14×	2.10×	2.08×	2.13×	2.30×
MR2	2.00	2.77×	3.28×	3.34×	3.31×	3.31×	3.35×	3.52×
MR3	0.50	3.47×	3.93×	4.09×	3.89×	3.84×	4.33×	4.51×
MR4	0.50	3.58×	4.12×	4.18×	4.09×	4.03×	4.21×	4.49×
NM1	0.50	4.88×	5.15×	5.02×	5.26×	5.28×	5.76×	6.28×
RG1	6.86	1.45×	1.70×	1.70×	1.70×	1.69×	1.63×	1.72×
RG2	7.18	3.68×	4.23×	4.32×	4.28×	4.30×	4.32×	4.51×
RG3	5.91	5.46×	6.08×	6.82×	6.11×	6.12×	6.99×	7.31×
SC1	9.71	3.46×	3.71×	3.70×	3.73×	3.74×	3.85×	4.73×
XA1	2.00	4.37×	5.01×	4.94×	5.04×	5.03×	4.88×	5.39×

TABLE VIII: Encoding throughput (MB/s), Intel Core Ultra 9 285K (AMD64). Bold = fastest per row.

Image	MIC-Go	MIC-4s-C	MIC-C	Wav+SIMD	HTJ2K	JPEG-LS	PICS-4	PICS-8
MR	121	290	273	85	177	71	186	128
CT	180	359	311	84	178	104	301	312
CR	233	461	423	120	193	89	732	1,102
XR	254	550	519	127	214	95	775	1,212
MG1	380	861	820	155	508	239	1,112	1,651
MG2	380	857	830	153	507	235	1,119	1,676
MG3	256	556	514	97	202	109	832	1,317
MG4	336	738	710	107	354	162	1,098	1,901
CT1	206	413	384	94	216	132	310	329
CT2	192	371	340	89	194	132	320	294
MG-N	254	562	529	99	202	107	842	1,353
MR1	221	460	429	101	195	92	364	347
MR2	275	566	532	107	230	102	643	857
MR3	300	609	591	119	292	142	498	448
MR4	249	494	451	108	226	142	370	368
NM1	237	467	444	117	242	132	360	302
RG1	229	485	397	123	198	70	651	942
RG2	280	582	488	125	254	112	842	1,220
RG3	275	550	512	128	273	143	863	1,222
SC1	286	577	551	83	235	169	924	1,425
XA1	242	496	471	101	219	114	616	832

Overall, the study indicates that 16-bit-native entropy coding is a useful design point for lossless medical image compression, particularly in applications where decompression throughput, implementation simplicity, and deployment portability are important.

Deployment guidance. For deployments where implementation simplicity is the priority, the pure Go or JavaScript decoder suffices. For maximum single-threaded native throughput on a server, use MIC-4state-C. For large images on multicore systems, PICS-C-8 provides the highest decompression rates on both ARM64 and AMD64.

Future work should strengthen the evidence base with larger datasets, shuffle-based general-purpose baselines, formal statistical reporting, and direct browser benchmarking.

MIC is open-source and available at <https://github.com/pappuks/medical-image-codec>.

the developers of OpenJPH and CharLS for their open-source implementations used in the comparative evaluation.

CONFLICT OF INTEREST

The author declares no competing financial interests or personal relationships that could have influenced the work reported in this paper.

DATA AVAILABILITY

The test images are drawn from publicly available DICOM datasets: NEMA WG-04 compression samples [31], NEMA 1997 CD, and the Clunie Breast Tomosynthesis Case 1 [32]. The MIC implementation, benchmark harness, and reproduction instructions are available at <https://github.com/pappuks/medical-image-codec>.

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TABLE IX: Encoding throughput (MB/s), Apple M2 Max (12-core ARM64), all variants. Bold = fastest per row.

Image	MIC-Go	MIC-4s	MIC-4s-C	MIC-C	Wav+SIMD	HTJ2K	JPEG-LS	PICS-2	PICS-4	PICS-8
MR	180	219	243	305	102	217	104	111	104	103
CT	208	222	314	332	89	258	166	195	358	313
CR	307	312	441	416	138	311	119	488	856	1,195
XR	336	322	498	500	126	269	123	411	738	1,136
MG1	512	508	621	651	162	676	320	858	1,437	2,001
MG2	517	505	644	702	160	700	315	829	1,372	2,095
MG3	355	352	469	425	95	293	153	530	862	1,491
MG4	458	463	554	503	106	451	234	654	1,186	1,984
CT1	292	267	383	388	89	314	204	299	397	347
CT2	228	236	356	335	95	294	231	286	373	343
MG-N	341	357	424	427	94	309	156	538	929	1,421
MR1	309	307	400	441	104	271	151	176	301	301
MR2	350	350	498	514	106	339	136	463	672	821
MR3	385	379	538	568	121	389	183	287	434	516
MR4	305	310	466	471	118	354	252	262	363	325
NM1	302	302	438	444	114	346	178	273	323	474
RG1	284	315	471	367	127	256	107	349	648	985
RG2	390	398	493	520	140	402	155	598	1,034	1,583
RG3	374	388	488	485	153	455	198	566	1,029	1,389
SC1	414	413	508	466	97	349	297	655	1,170	1,712
XA1	335	324	449	463	111	357	154	437	708	944

TABLE X: Decompression throughput (MB/s), Apple M2 Max (ARM64). Bold = fastest per row (single-threaded: first 5 cols; PICS: last 3 cols).

Image	MIC-Go	MIC-4s-C	Wav+SIMD	HTJ2K	JPEG-LS	PICS-C-2	PICS-C-4	PICS-C-8
MR	144	348	248	265	102	530	710[†]	482
CT	191	356	316	307	137	524	955	1,092
CR	296	524	567	367	153	867	1,635	2,661
XR	308	533	627	334	108	874	1,666	3,025
MG1	482	683	678	810	409	1,205	2,112	3,656
MG2	479	686	697	790	416	1,225	2,120	3,773
MG3	308	531	422	338	153	864	1,673	3,117
MG4	417	625	516	548	184	1,093	2,004	3,689
CT1	239	436	425	362	182	686	1,013	1,183
CT2	238	439	481	375	175	676	1,041	1,189
MG-N	316	536	468	340	153	883	1,711	3,175
MR1	278	521	435	325	116	751	1,207	1,402
MR2	333	563	498	388	172	913	1,552	2,466
MR3	375	639	507	441	236	908	1,430	1,614
MR4	316	571	479	406	197	818	1,341	1,558
NM1	327	632	575	410	210	888	1,400	1,679
RG1	235	406	584	332	104	602	1,128	2,017
RG2	367	590	644	443	193	986	1,803	3,194
RG3	374	604	656	562	246	1,035	1,944	3,302
SC1	375	587	388	401	229	1,017	1,861	3,279
XA1	331	576	459	419	204	928	1,583	2,493

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TABLE XI: Decompression throughput (MB/s), Intel Core Ultra 9 285K (AMD64). Bold = fastest per row within single-threaded and PICS groups.

Image	MIC-Go	MIC-4s-C	MIC-4s-SIMD	Wav+SIMD	HTJ2K	PICS-C-4	PICS-C-8
MR	251	501	708	194	570	301	124 [†]
CT	231	403	487	364	544	676	339
CR	324	599	744	723	708	1,738	2,435
XR	363	601	803	681	570	1,714	1,994
MG1	617	789	1,119	746	1,235	1,872	2,514
MG2	562	800	1,166	848	1,297	2,244	2,085
MG3	357	633	669	678	644	1,823	2,538
MG4	523	743	773	808	916	2,124	2,707
CT1	322	520	676	525	657	857	423
CT2	293	514	636	705	627	847	687
MG-N	368	635	669	705	643	1,872	2,191
MR1	356	609	766	532	654	945	366
MR2	352	658	975	601	749	1,121	793
MR3	450	728	809	676	802	920	705
MR4	390	596	861	738	660	493	701
NM1	367	717	668	715	627	783	405
RG1	302	497	593	705	557	1,248	1,494
RG2	433	676	861	811	823	1,841	1,912
RG3	460	706	858	881	894	1,969	1,613
SC1	464	695	888	710	728	1,819	1,889
XA1	413	693	836	538	797	1,173	1,513

TABLE XII: FSE decompression throughput (MB/s) for representative modalities. ARM64: Apple M2 Max, pure Go. AMD64: Intel Core Ultra 9 285K, 4-state uses BMI2 assembly (fse4StateDecompKernel).

Image	ARM64 (MB/s)			AMD64 (MB/s)		
	1-st	4-st	gain	1-st	4-st	gain
MR	514	841	+64%	600	828	+38%
CT	829	1,295	+56%	941	1,631	+73%
CR	140	966	+590%*	1,994	4,535	+127%
XR	1,444	1,952	+35%	3,195	5,771	+81%
MG1	1,140	1,873	+64%	3,379	4,952	+47%
MG-N	2,944	5,696	+94%	2,722	4,981	+83%
NM1	1,658	2,254	+36%	2,078	2,407	+16%
RG1	1,676	3,128	+87%	2,506	4,336	+73%
Geomean (21)	1,418	2,375	+68%	2,309	3,306	+43%

TABLE XIII: JavaScript decoder throughput (4-state), Node.js v24.8, Apple M2 Max. Median over 20 iterations.

Image	Dimensions	Ratio	ms	MB/s
MR	256×256	2.35×	3.0	42
CT	512×512	2.24×	13.5	37
CR	2140×1760	3.69×	161.2	45
MG1	2457×1996	8.79×	80.9	116
MG3	4774×3064	2.29×	638.4	44

TABLE XIV: PICS parallel JavaScript decoder (worker_threads), Apple M2 Max. Speedup relative to 1 worker.

Image	Strips	ms	MB/s	Speedup
MR	4	1.3	94	2.57×
CT	4	5.0	101	2.95×
CR	8	31.7	227	5.53×
MG1	8	19.4	483	4.36×

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